

Basketball Royale

Game design document



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Introduction

Basketball Royale is a game made exclusively for Twitch; it will be used as a Twitch extension. The streamer controls the game and the viewers have limited control. Through chat commands like !start and !reset, streamers can control the game. Viewers can play by using chat commands too like !join.

The game is meant as a quick game giving the broadcaster time to go to the bathroom, answer the door, etc. It can also be used in giveaways to determine the winner(s).

Gameplay

Basketball Royale has two game modes: Royale and Arcade. In both cases, the game starts when the chat command !start is used by the broadcaster. In the following 60 seconds, viewers can join the game by typing !join in the chat. After the join period, there are 5 seconds to let the players know the game is starting, then the game begins.

In Royale mode, players can score points by throwing basketballs in the moving basket. The game will automatically throw basketballs to the moving basket. Every player has one ball thrown for them. Without viewer chat commands, this game mode is very suitable for large streams.

In Arcade mode, basketballs that go through the net give points. The player has to aim and throw the balls themselves by using !throw 30 50 for example.

After a round, the players who are in the top qualify to the next round, see [values](#) for the percentage that qualify. Each round lasts a set time, also specified in [values](#) and has special events listed in [special events](#), depending on how many rounds have passed, e.g. round 6 has more events than round 1. Eventually, one player is left who wins the game. The game displays the winner and in Arcade mode the final scoreboard. See Figure 1.



Figure 1: Timer, winner and scoreboard.

Unique selling points

Viewer interaction:

- Viewers of the (live) broadcast can play the game.

Viewer recognition:

- When a player scores, their name is displayed briefly on the scoreboard.
- The winning player's name is displayed at the end of the game.

Game flow

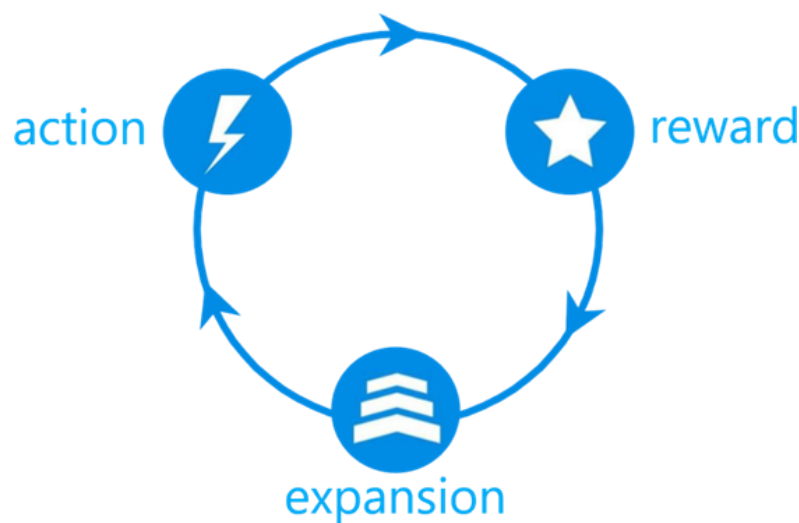


Figure 2: Game core loop (Zuza, n.d.).

The game flow seen in Figure 2 is composed of the following components:

1. Action: What does the player do?
The player throws balls in the basket.
2. Reward: What does the player receive for the action?
Royale mode: The player qualifies for the next round.
Arcade mode: Points are awarded to the player.
3. Expansion: What can the player do with the reward?
Royale mode: The player can play the next round, and is closer to winning.
Arcade mode: The more points the player has, the higher the chance they will get to the next round and win the game.

Primary game flow

Action: The player throws balls in the basket.

Reward: Points are awarded to the player or the player qualifies for the next round.

Expansion: In Arcade mode, the more points the player has, the higher the chance they will get to the next round. In both modes, the player is closer to winning the game.

Secondary game flow

Action: The game is played.
Reward: The owner of the game is given more special events.
Expansion: The granted special events can occur in the next games.

Controls

Broadcaster

!pause / !unpause
Immediately (un)pauses the current round.

!start
Starts the join phase where viewers can join the game.

!reset
Immediately resets the game.

!stop
Immediately stops the whole game and makes the game (source) invisible.

Viewer

!join
The viewer is added as a participant to the game.

Participant (Arcade mode only)

!throw {x} {y}
A basketball is thrown towards the specified coordinates. X and Y are values between 0 and 100. 0 0 is the bottom left, 100 100 is the top right as seen in Figure 3.

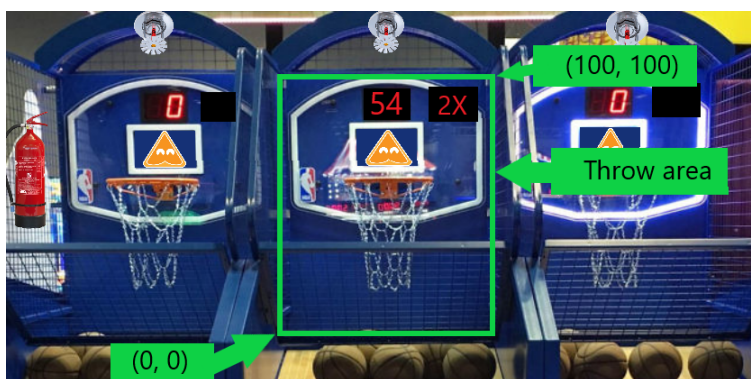


Figure 3: Throw area.

Main gameplay concepts

Genre

The game is a **battle royale** with rounds, where each round, players are disqualified until there is one winner left.

Setting

As shown in Figure 4, the game is set in an arcade, where the players are at a basketball arcade machine. The environment is filled with other arcade machines and objects that would naturally be in an arcade like fire sprinklers, flashing lights, and a fire extinguisher. The sound includes noises made by other arcade machines, people in the arcade making noise and sounds made by the basketball arcade machine. The basketball arcade machine itself makes sounds like an airhorn when the game begins, the sounds of basketballs bouncing on various materials, a buzzer when the game ends, and the celebration of a winner.

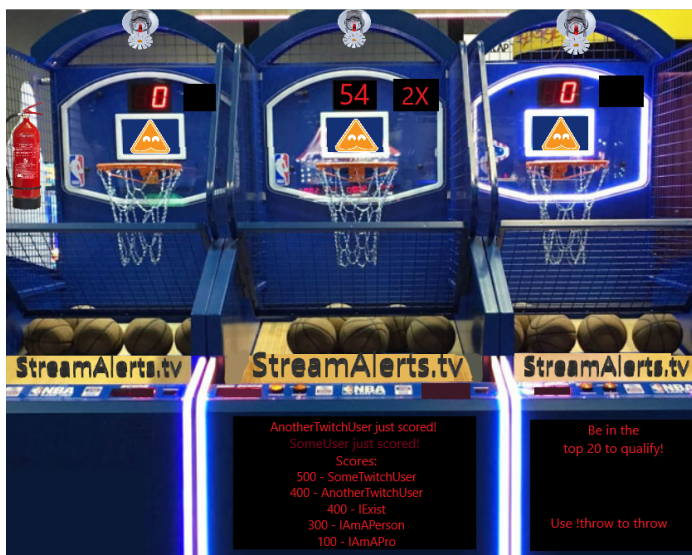


Figure 4: Setting.

Game world

The player is unable to move because the focus of the game is on the basketball arcade machine and there are limitations because the game is developed for Twitch, where participants have limited controls and the broadcaster does not have a need to move in the game. The more rounds are played, the more events happen, described in [special events](#).

Mood

The game is casual, but chaotic. The game has a set structure where players know that there are rounds of a set duration, whereafter they either qualify or not. The setting however includes flashing colors, electronic sounds, and the unexpected events (see [special events](#)).

Theme

Competition is the theme of the game, as all players are competing to win the rounds and become the winner of the game.

Mechanics

- Starting the game.
- Resetting the round.
- Stopping the game.
- Joining the game.
- Setting the position of the throw for basketballs
- Setting the strength of the throw for basketballs.

Dynamics

- Throwing basketballs.
The throw of a basketball consists of a position and strength. When the position is in the middle, the strength does not need to be very high. When the position is far to the left or right, the strength needs to be much higher, as the basketball needs to travel a longer distance.
- Timing throws (Arcade mode only).
When throwing balls simultaneously, a percentage depending on the amount of participants, seen in [values](#). Simultaneously thrown balls hit each other and they miss. In Royale mode, this is all based on chance, because the players have no way to control the timing.
- Aiming and anticipating where the basket will be (Arcade mode only).
Since the basket is moving along the paths shown in Figure 5, 6, and 7, and the player is not given the exact coordinates of the basket, the players need to aim where they want to throw the basketballs and need to take the movement of the basket into account.



Figure 5, 6, and 7 Paths of the basket.

Aesthetics

- Upbeat, fast paces, arcade music.
- A wide variety of bright colors.
- Flashing lights.

Special events

Fire sprinkler

The fire sprinkler goes off and the arcade becomes flooded, causing balls to float upward.

See Figures 8 and 9.



Figure 8: Sprinkler positions.



Figure 9: Sprinkler effect.

Fire extinguisher

A fire extinguisher, see Figure 10, is hit by a basketball and explodes filling the room with fog, making it hard to see as seen in Figure 11.

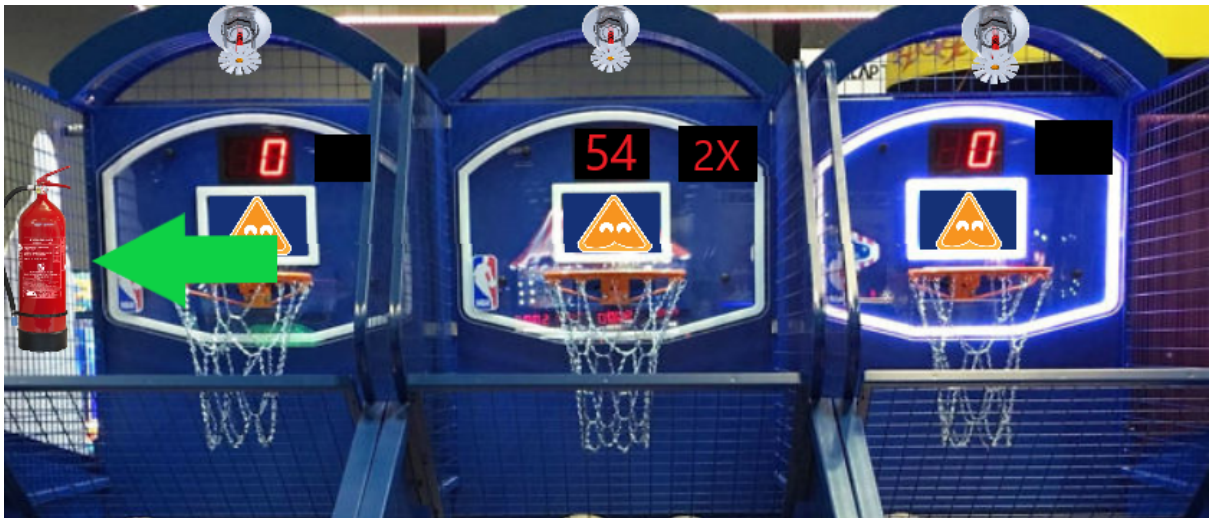


Figure 10: Fire extinguisher position.



Figure 11: Fog effect.

Lights out

The light went out, causing no 'global' illumination. Only the lights of the arcade machines are visible. See Figure 12.

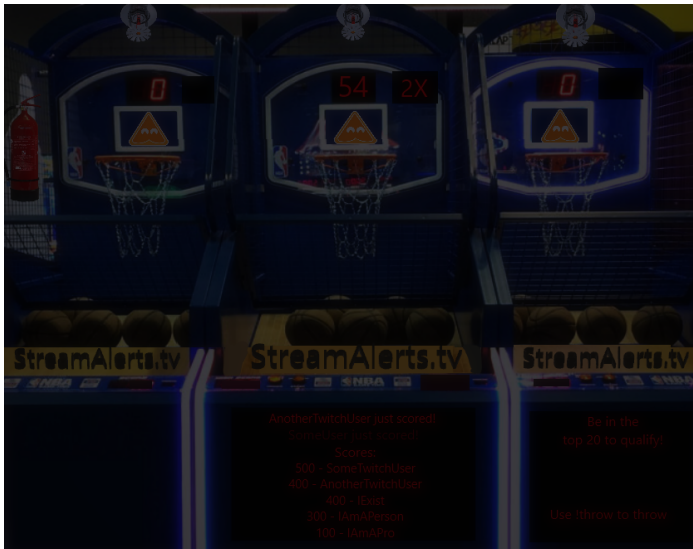


Figure 12: Light effect.

Multiplier (Arcade mode only)

At random times, a multiplier is added as seen in Figure 13, making points of a score worth more for a certain period, see [values](#). However, this means that more players will throw balls in this period, causing them to collide and miss.

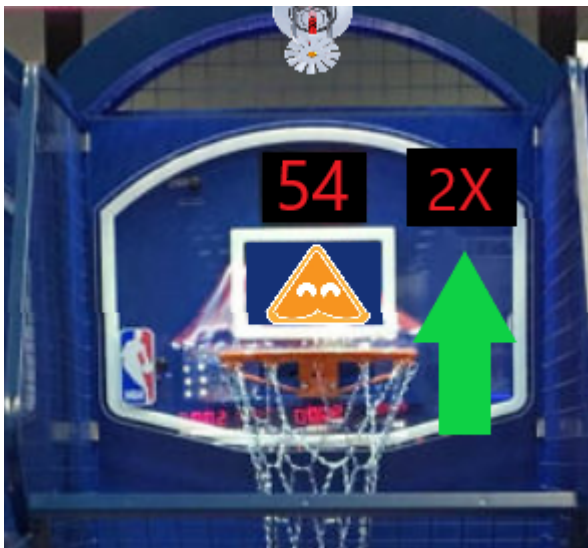


Figure 13: Multiplier.

Bits (Arcade mode only)

When bits are spent at the broadcast, the points of a score of the participant are increased for a duration based on the amount of bits. Can be disabled in the settings.

Subscription (Arcade mode only)

When a participant subscribes to the broadcaster, their points are worth more during that round. Can be disabled in the settings.

Design



Figure 14: Design.

The area of focus is set in the middle of the screen for ease indicated with a green box in Figure 14. This way, everything happening outside the area of focus is close within the peripheral vision.

The basket is the most important object alongside the balls themselves. Therefore it is located in the middle of the screen. It will be moving inside the area indicated with an orange box so it will stay in the middle.

Basketballs will be thrown from the bottom of the screen, making it look like the balls are thrown from the players themselves, improving the simulation experience.

The scoreboard is an important panel that needs to be visible at all times and is rather large due to all the scores being displayed. To separate the action from the scoreboard, it is put in the bottom of the screen. It could also be between the timer and the sprinklers.

The timer is a crucial component of the game, so it is placed near the top, where the balls are going through the basket.

The multiplier is placed next to the timer, because it is an element that influences the game heavily and is not screen filling. The attention of the players will be the scoreboard and the timer. A multiplier integrated with the scoreboard screen could be too subtle.

Sprinklers are placed on the roof, as usual buildings have.

The fire extinguisher is placed left of the arcade machines, as it would not make sense to place it **on** the machines. It could also be placed right of the arcade machines.

The StreamAlerts.tv link is put on the area where balls roll back so it is between the throw area and the scoreboard. Because it is in the area of focus, players see it the most times.

The logo of Stream Alerts is put on the basket so there is always a reference that does not require reading to Stream Alerts in the area of focus.

In the bottom right, instructions are shown. This region stands out in between games as it is relatively empty, and the letters are in a contradicting color, making them stand out. When in the join period, it will display how to join the game. During the rest of the round, it will show when players qualify and how they can throw. This adds realism as the display (of the right arcade machine) is not very actively 'getting input' from the main arcade machine.

The left bottom and right middle part of the screen are left empty for streamers to place their webcam, chat, etc.

Values

Description	Value
Round length	1 minute
Percentage that qualifies	50%
Percentage of players that throw at the same time causes a collision	5% or higher starting at 40 players, otherwise 100%
Amount of shots available per round in Arcade mode	5
Points per score in Arcade mode	100
Multipliers in Arcade mode	1.5, 2
Event frequency round 1	No events
Event frequency round 2	15 - 25 seconds

Event frequency round 3 and above	10 - 20 seconds
Event frequency last round	No events

Settings

Join period:

The period in which viewers are able to use !join.

Bits give increased points in Arcade mode:

See [special events](#).

Subscriptions give increased points in Arcade mode:

See [special events](#).

Branding

All arcade machines have the brand of Stream Alerts on them, with StreamAlerts.tv at the bottom. See Figure 15.



Figure 15: Branding.

Potential problems

Slow mode

Twitch has a moderation possibility that allows broadcasters to add a cooldown on messages. A user cannot send any message until that cooldown has finished. By default this is 30 seconds.

Unique messages

Twitch forbids identical messages sent consecutively within 30 seconds.

Displaying all basketballs

With 10 people playing, all basketballs could easily be displayed. However, if there are 10,000 players, this would fill the screen easily. There must be a limit on how many basketballs are shown at any time.

Possible solutions

Slow mode / unique messages

Throw strength (discarded)

Participants have to throw using `!throw {x}`, where `x` is the strength of the throw. To prevent repeating messages once the right strength is figured out, the basket has to be moving or the required strength has to vary in another way. The strength is a value between 0 and 100.

Throw coordinates

As an advancement on strength, the player has to specify coordinations, using `!throw {x} {y}`, where `x` is the horizontal value and `y` the vertical value. This gives the player more control at the cost of complexity, also making it harder to figure out the right values. The basket has to be moving or the required coordinates have to vary in another way in order to prevent repeating messages. The coordinates are values between 0 and 100 in both directions.

Displaying all basketballs

Putting a hardcoded limit on the number of basketballs visible on the screen can be a suitable solution. When players score, it needs to be visible too, in the form of the name showing up on the screen. Making the limit too low makes it obvious that not all balls from players that scored were going in the basket. Making the limit too high makes the screen too crowded. The limit is set in [values](#).

Similar products

Stream Avatars' Basketball (Figure 16).



Figure 16: Stream Avatars' Basketball (Miller, n.d.).

Literature list

Miller, T. (n.d.). *Stream Avatars*. Stream Avatars. Retrieved February 08, 2023, from <https://www.streamavatars.com/>

Zuza, D. (n.d.). *Improve your game core loop*. CITYOS. Retrieved February 08, 2023, from <https://cityos.io/tutorial/634/Improve-Your-Core-Loop>